CONFIDENCE GATE: A FIXED PROBABILITY THRESHOLD FOR ACCEPTING CLASSIFIER PREDICTIONS

Anonymous authors
Paper under double-blind review

ABSTRACT

This manuscript is a synthetic skill-evaluation draft, not a report of completed research. We specify Confidence Gate, a rule applied to the predictions of a fixed classifier: a prediction is accepted when its maximum class probability is at least 0.8 and rejected otherwise. The rule is defined without modifying the classifier. The supplied material contains no dataset, named model, experimental measurements, implementation, related literature, or theoretical result. Accordingly, this draft formalizes the decision rule and identifies the information needed for evaluation, without claiming improved performance, novelty, or a correctness guarantee. The threshold is a supplied design choice whose empirical justification remains unavailable. All research conclusions beyond the stated rule require additional author evidence and review.

1 SCOPE AND PROBLEM

A classifier prediction may be passed through an acceptance rule that determines whether the prediction is retained or rejected. Confidence Gate uses only the largest class probability reported for that prediction. The supplied proposal sets the threshold to 0.8 and keeps the classifier fixed. This manuscript specifies that behavior; it does not establish whether the rule is useful on any particular task.

Draft status. This is a synthetic skill evaluation rather than a real research paper. No experiments or theoretical guarantees are available, and the authors have not reviewed the AI-assisted initial draft. The term "Confidence Gate" preserves the supplied method name. Here, "confidence" denotes the maximum reported class probability; it does not assert that this probability is calibrated or equals the probability that the prediction is correct.

AUTHOR INPUT PENDING: Provide the intended application, the task-specific reason for rejecting predictions, and evidence of the problem that the rule is intended to address. Novelty and comparison with existing approaches cannot be assessed without verified related literature.

2 DECISION RULE

Let x denote an input. Write $p_k(x)$ for the class probability reported by the fixed classifier for class $k \in \{1, \dots, K\}$, and let $\widehat{y}(x)$ denote its original predicted label. This notation represents the supplied description of class probabilities; it does not specify a model architecture or a training procedure. Define the score and acceptance indicator as

$$s(x) = \max_{1 \le k \le K} p_k(x), \qquad a(x) = \mathbf{1}\{s(x) \ge 0.8\}. \tag{1}$$

Confidence Gate returns the original prediction when $a(x) = 1$ and a rejection symbol $\perp$ when $a(x) = 0$:

$$g(x) = \begin{cases} \widehat{y}(x), & s(x) \ge 0.8, \\ \perp, & s(x) < 0.8. \end{cases} \tag{2}$$

In particular, a score exactly equal to 0.8 is accepted. Rejection provides no alternative label in the supplied rule. The classifier's original prediction is retained, so the gate does not introduce a

separate label-selection or tie-breaking procedure. How the classifier produces its probabilities and predicted labels is not specified.

Operationally, the rule takes the fixed classifier's reported probabilities and prediction, computes the maximum probability, compares it with 0.8, and either passes through the prediction or returns $\perp$. The material supplies no rationale for selecting 0.8 and no threshold-selection experiment. The value is therefore a fixed specification, not an empirically optimized parameter. **AUTHOR INPUT PENDING:** Specify the classifier, class set, probability-generation procedure, handling of invalid or missing outputs, numerical representation, and downstream treatment of rejection.

3 Evidence Status and Evaluation Requirements

No evaluation has been performed or reported in the supplied material. Table 1 records the missing inputs. In particular, an accepted prediction has a reported score of at least 0.8 by definition, but that rule alone does not establish an accuracy level, an error reduction, or any other performance benefit. No theorem or proof is asserted here.

Table 1: Material gaps. These entries describe availability, not experimental results.

Required information	Status and consequence
Dataset and task	Not supplied; task scope and data distribution are unknown.
Classifier	Specific model and training details are not supplied.
Evaluation protocol	Splits, seeds, comparison methods, and metric definitions are not supplied.
Measurements	No performance values or uncertainty estimates are supplied.
Code and resources	No code, execution environment, or resource measurements are supplied.
Related literature	No verified references are supplied; positioning is pending.
Theory	No theorem or proof is supplied; no guarantee is claimed.

An author-designed evaluation could examine how frequently the rule accepts predictions and how often those accepted predictions are wrong. For clarity, if labeled evaluation examples $(x_i, y_i)_{i=1}^n$ become available, one possible reporting convention is empirical acceptance frequency

$$\widehat{C} = \frac{1}{n} \sum_{i=1}^{n} a(x_i), \tag{3}$$

and accepted-prediction error

$$\widehat{R} = \frac{\sum_{i=1}^{n} a(x_i) \mathbf{1}\{\widehat{y}(x_i) \neq y_i\}}{\sum_{i=1}^{n} a(x_i)}, \quad \text{when } \sum_{i=1}^{n} a(x_i) > 0. \tag{4}$$

If no prediction is accepted, the latter quantity is undefined and should be reported as such. These are proposed descriptive reporting definitions, not measurements, author-approved study choices, or a theoretical guarantee. Any comparison must also report acceptance frequency so that error figures can be interpreted with their retained sample size. The authors must decide the protocol and appropriate comparisons for the actual task.

AUTHOR INPUT PENDING: Supply actual data, model details, evaluation choices, code, seeds where applicable, results, and the basis for uncertainty reporting. A prospective evaluation should distinguish the specified threshold from any subsequent threshold tuning. No claim of improvement can be added until its comparison conditions and evidence are available.

4 Related Work, Limitations, and Conclusion

AUTHOR INPUT PENDING: Provide and verify the closest scholarly references, then compare their assumptions, acceptance rules, and supported conclusions with this proposal. No novelty claim is justified by the current material.

The current material supports the definition of Confidence Gate and its inclusive threshold boundary. It does not support a claim about robustness, calibration, fairness, computational savings, or deployment suitability. The consequences of rejecting an input depend on an unspecified downstream procedure. There is also no empirical basis for recommending 0.8 across classifiers, datasets, or applications.

Confidence Gate accepts a fixed classifier's prediction precisely when its maximum reported class probability is at least 0.8. This draft makes that rule explicit and exposes the missing evidence. Assessing its value requires author-reviewed research materials; the draft is not ready for submission.

AI USE STATEMENT

AI was used to organize this initial draft and express the supplied material in English. The authors have not yet reviewed the resulting text. No experimental results, references, theorems, or proofs were supplied for this drafting task; this manuscript does not claim that AI use established any research result. **AUTHOR INPUT PENDING:** The authors must verify the manuscript and update this statement to reflect the complete, actual AI-use record before any submission.

REPRODUCIBILITY STATEMENT

The threshold rule in Section 2 is specified, but the classifier, data, implementation, and evaluation protocol are unavailable. No executable research implementation or experimental reproduction is provided. The supplied material is insufficient to reproduce a performance claim because no such claim has been verified. The accompanying `references.bib` is deliberately empty: no verified scholarly references were supplied.